Incorporating Geometric Phase Effects by Local Diabatic Representation

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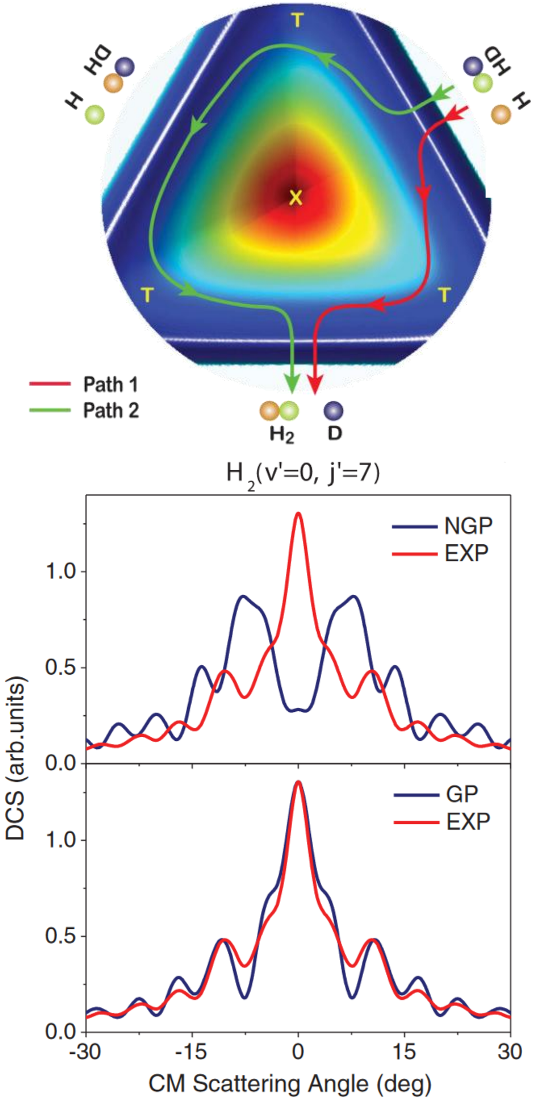
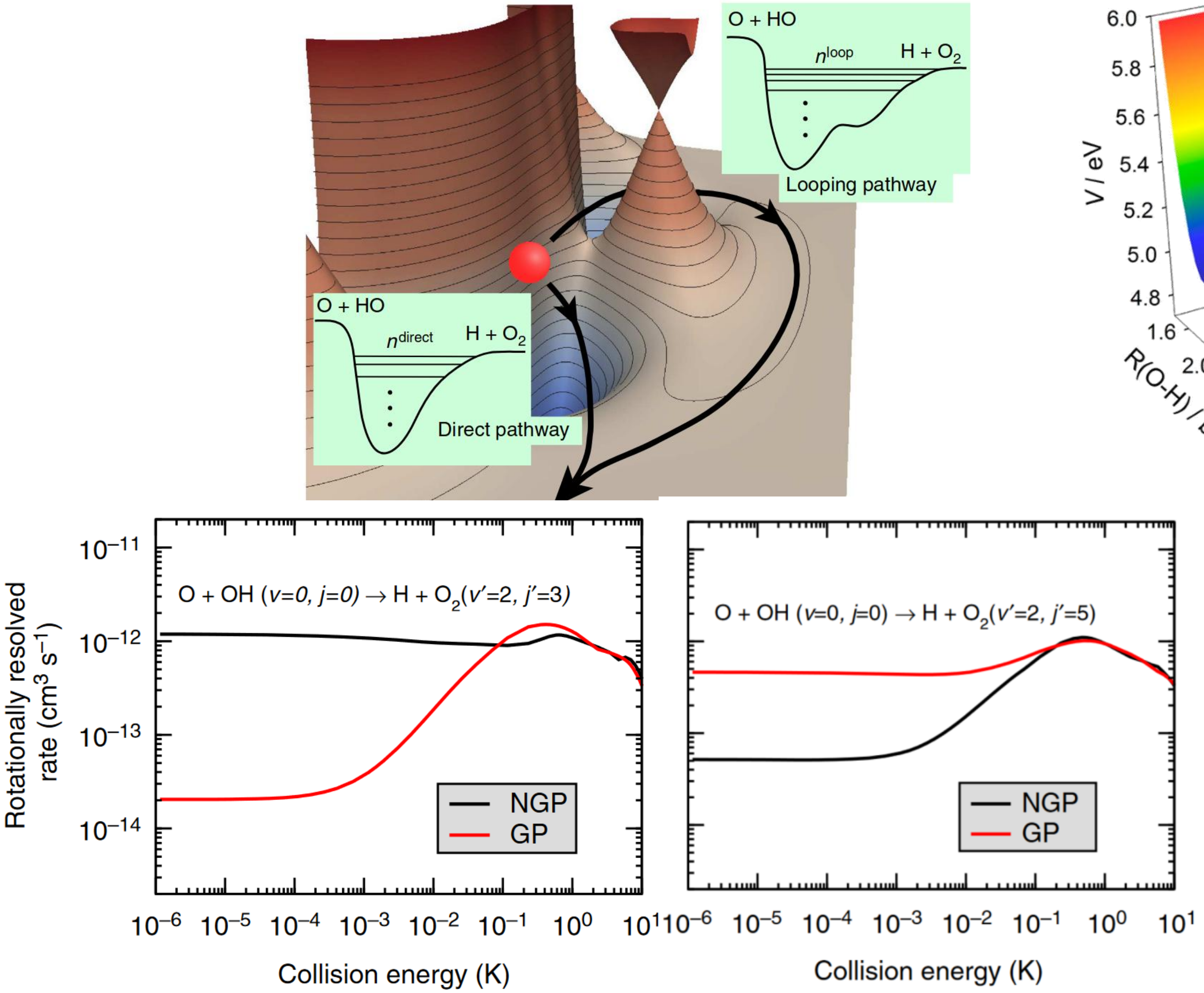
Geometric phase effects profoundly impact reactive dynamics.

Enhancing the forward scattering



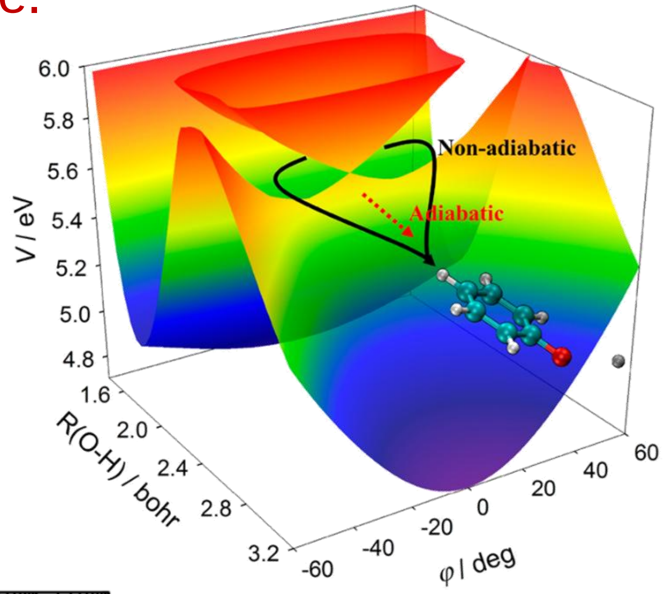
Yuan et al., Nat. Commun. 2020, 11, 3640

Suppressing or enhancing the reaction rates by nearly two orders of magnitude.



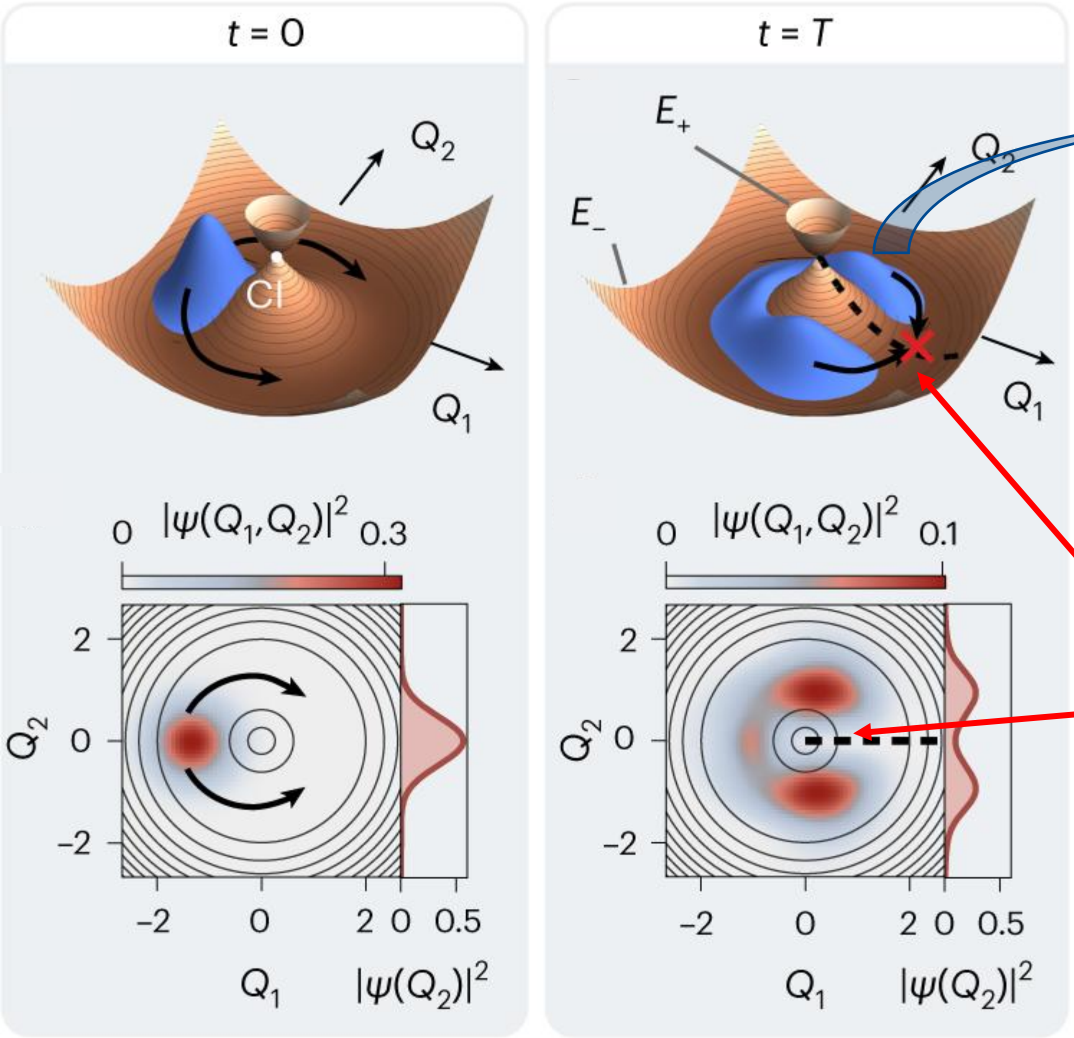
Kendrick et al., Nat. Commun., 2015, 6, 7918

100 times shorter



Xie et al. Acc. Chem. Res. 2019, 52, 501−509

The geometric phase effect arises from the sign change of the adiabatic electronic wave function along a closed loop around a conical intersection.



$$e^{i\pi} = -1$$

$$\underset{\text{single valued}}{\boxed{\Psi(\mathbf{r}, \mathbf{R}, t)}} = \sum_{\alpha} \underset{\text{change sign}}{\boxed{\phi_{\alpha}(\mathbf{r}; \mathbf{R})}}\chi_{\alpha}(\mathbf{R}, t)$$

Destructive interference of nuclear wave packet.

C. H. Valahu, et al. Nat. Chem. 2023, 15, 1503–1508

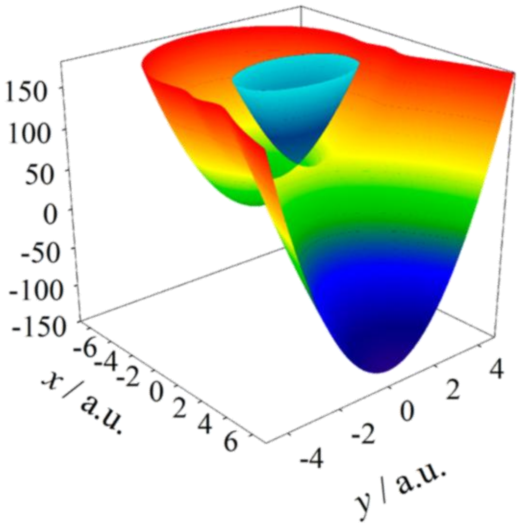
Challenges of conventional approaches to capture geometric phase effect.

- adiabatic representation:

$$\Psi_N^{\mathrm{total}}(\mathbf{r},\ \mathbf{R}) = \sum_{n=1}^{N^{\mathrm{state}}} \tilde{\Phi}_n^{(a)}(\mathbf{r};\ \mathbf{R})\Theta_{N,n}^{(a)}(\mathbf{R})$$

$$\tilde{\Phi}_n^{(a)}(\mathbf{r};\ \mathbf{R}) = e^{iA_n(\mathbf{R})}\Phi_n^{(a)}(\mathbf{r};\ \mathbf{R})$$

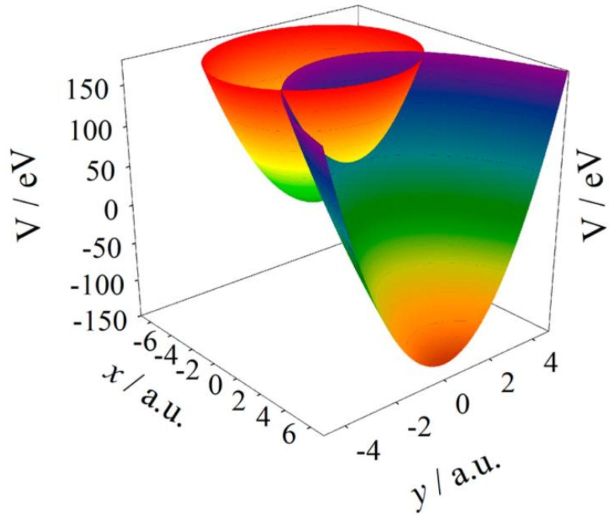
Vector potential: $A_n(\mathbf{R})$



- Divergence of vector potential and derivative coupling
- Locate conical intersection seam

- Diabatic representation:

$$\Psi_N^{\mathrm{total}}(\mathbf{r},\ \mathbf{R}) = \sum_{m} \Phi_m^{(d)}(\mathbf{r})\Theta_{N,m}^{(d)}(\mathbf{R})$$



- Numerically infeasible for multi-atomic systems

Xie et al., J. Chem. Theory Comput. 2017, 13, 1902−1910
Mead et al., J. Chem. Phys. 1979, 70, 2284−2296

The local diabatic representation (LDR) provides a divergence-free framework for capturing geometric phase effect.

- LDR

$$\Psi(\mathbf{r}, \mathbf{R}, t) = \sum_n \sum_\alpha C_{n\alpha}(t)\phi_\alpha(\mathbf{r}; \boxed{\mathbf{R}_n})\chi_n(\mathbf{R}; \mathbf{R}_n)$$

- Born-Huang expansion

$$\Psi(\mathbf{r}, \mathbf{R}, t) = \sum_\alpha \phi_\alpha(\mathbf{r}; \boxed{\mathbf{R}})\chi_\alpha(\mathbf{R}, t)$$

$\phi_\alpha(\mathbf{r}; \mathbf{R}_n)$: **adiabatic** electronic states at $\mathbf{R}_n$
remove the nuclear dependence of electronic states

- equation of motion: **full quantum**

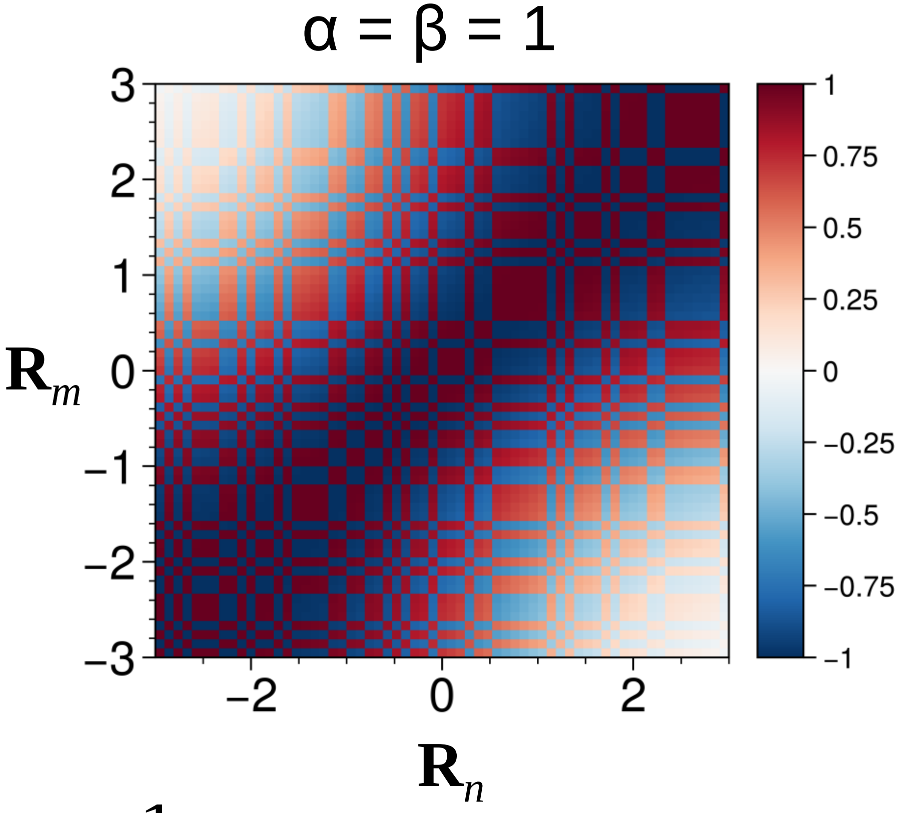
$$i\dot{C}_{m\beta}(t) = E_\beta(\mathbf{R}_m)C_{m\beta}(t) + \sum_{n,\alpha} T_{mn}\boxed{A_{m\beta,n\alpha}}C_{n\alpha}(t)$$

$A_{m\beta,n\alpha}$: **Divergence free.** Always bounded [-1,1] , even at intersections

Geometric phase effect is encoded in the electronic overlap matrix.

$$A_{m\beta,n\alpha} = \langle \phi_\beta(\mathbf{R}_m) | \phi_\alpha(\mathbf{R}_n) \rangle_{\mathbf{r}}$$

- **Geometric phase** → Phase of intrastate overlap matrix $\mathbf{A}_{m\alpha,n\alpha}$



- **LDR and BO dynamics**

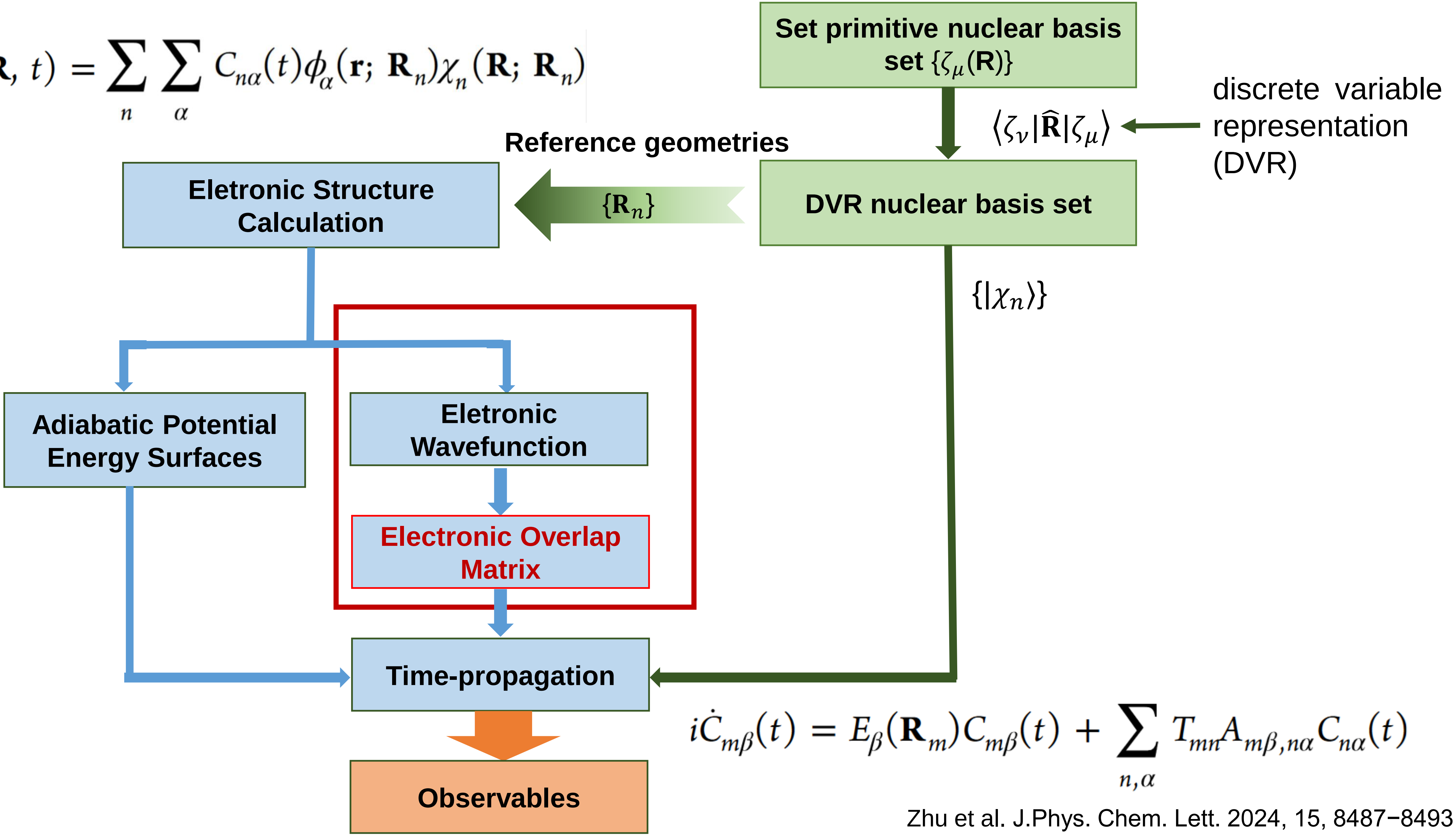
$$A_{m\beta,n\alpha} = \langle \phi_\beta(\mathbf{R}_m) | \phi_\alpha(\mathbf{R}_n) \rangle_{\mathbf{r}} \qquad A_{m\beta,n\alpha} = 1$$

Topology of electronic states → Born-Oppenheimer dynamics

Yujuan Xie, Bing Gu, arXiv:2501.05003

Workflow of ab initio LDR

$$\Psi(\mathbf{r}, \mathbf{R}, t) = \sum_{n} \sum_{\alpha} C_{n\alpha}(t)\phi_{\alpha}(\mathbf{r}; \mathbf{R}_n)\chi_n(\mathbf{R}; \mathbf{R}_n)$$



$$i\dot{C}_{m\beta}(t) = E_\beta(\mathbf{R}_m)C_{m\beta}(t) + \sum_{n,\alpha} T_{mn}A_{m\beta,n\alpha}C_{n\alpha}(t)$$

Zhu et al. J.Phys. Chem. Lett. 2024, 15, 8487−8493

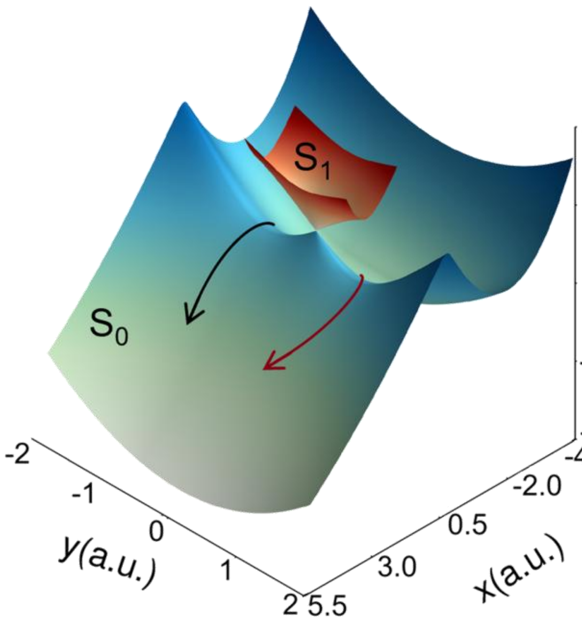
Geometric phase captured with only a single state.

- Two-state vibronic model
- Single-state LDR

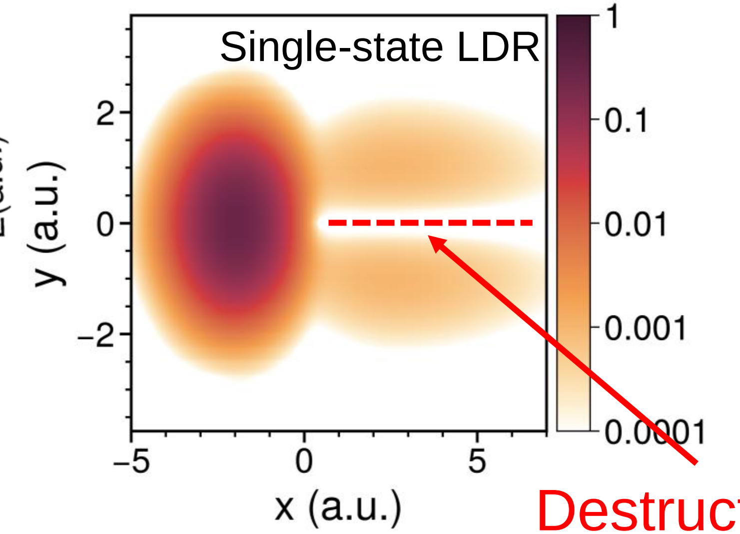
$$\Psi(\mathbf{r},\mathbf{R},t)=\sum_n C_n(t)\psi(\mathbf{r};\mathbf{R}_n)\chi_n(\mathbf{R})$$

$$A_{mn}=\langle\psi(\mathbf{R}_m)|\psi(\mathbf{R}_n)\rangle_{\mathbf{r}}$$

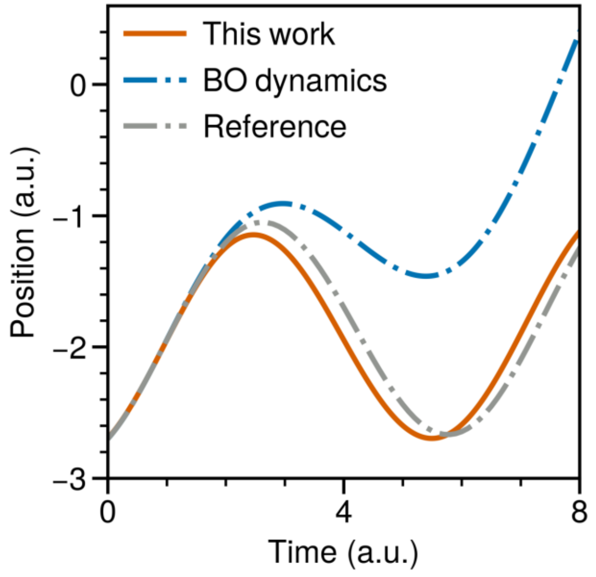
- geometric phase manifested as a nodal line along the $y = 0$.
- The reaction rate increases in the absence of the geometric phase effect.

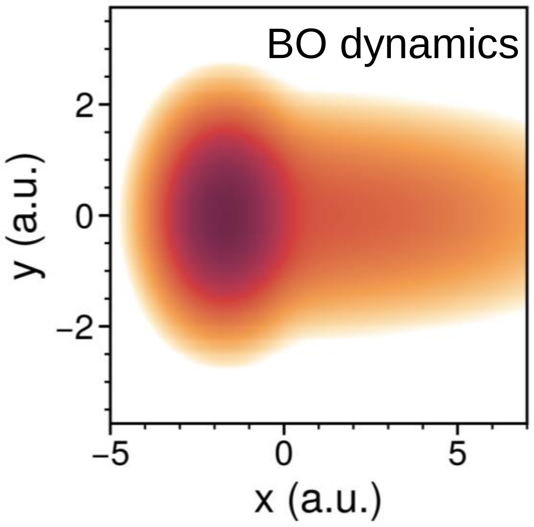


Ground state only



Destructive interference

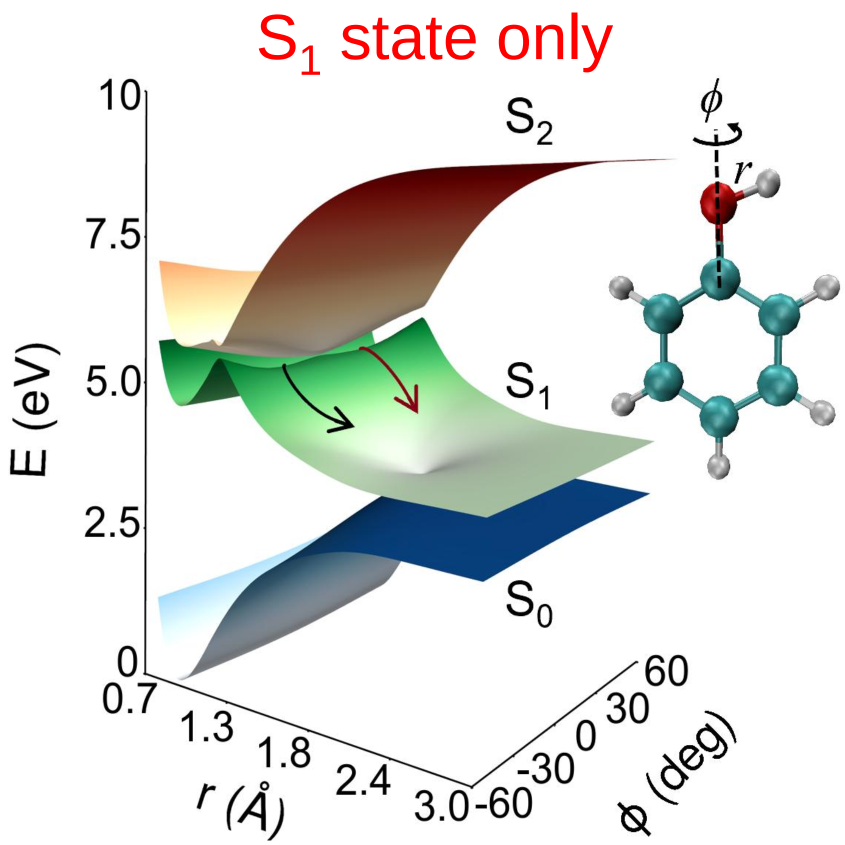


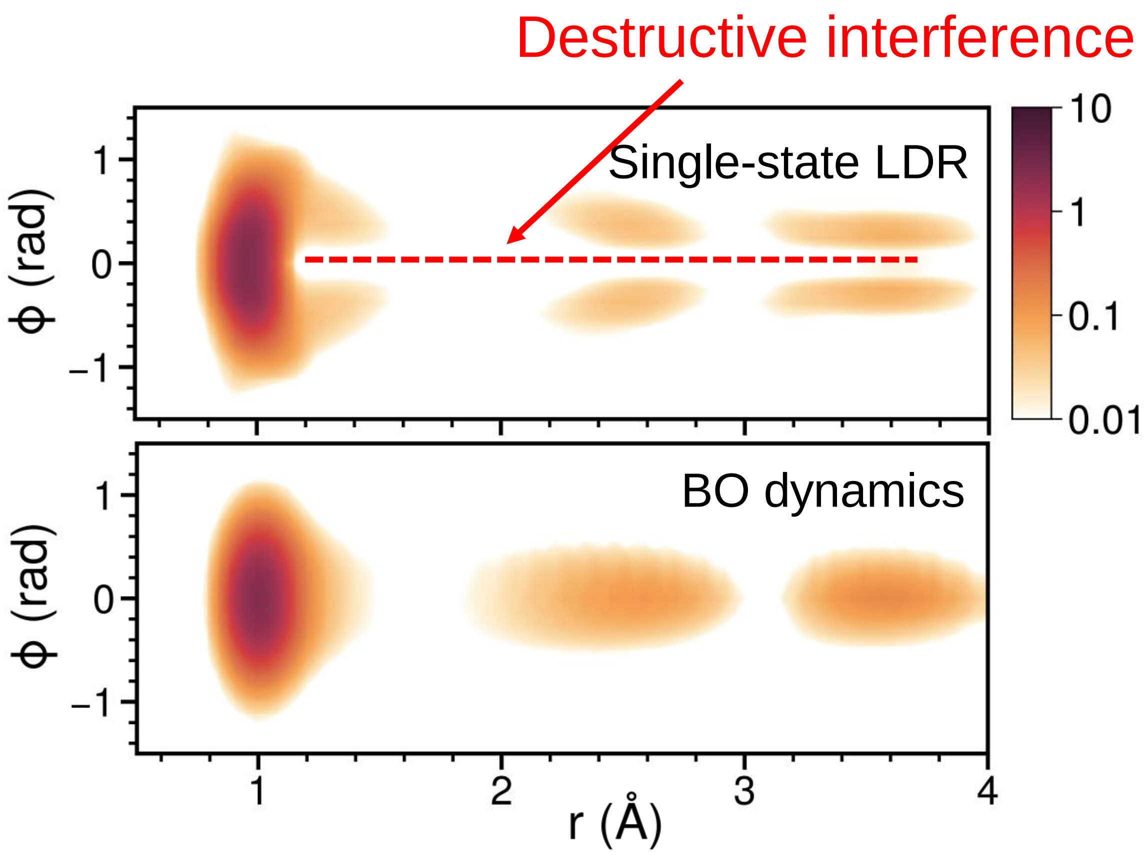


Xie et al., Phys. Rev. A, 95, 022104 (2017)
In preparation

Geometric phase captured with only a single state.

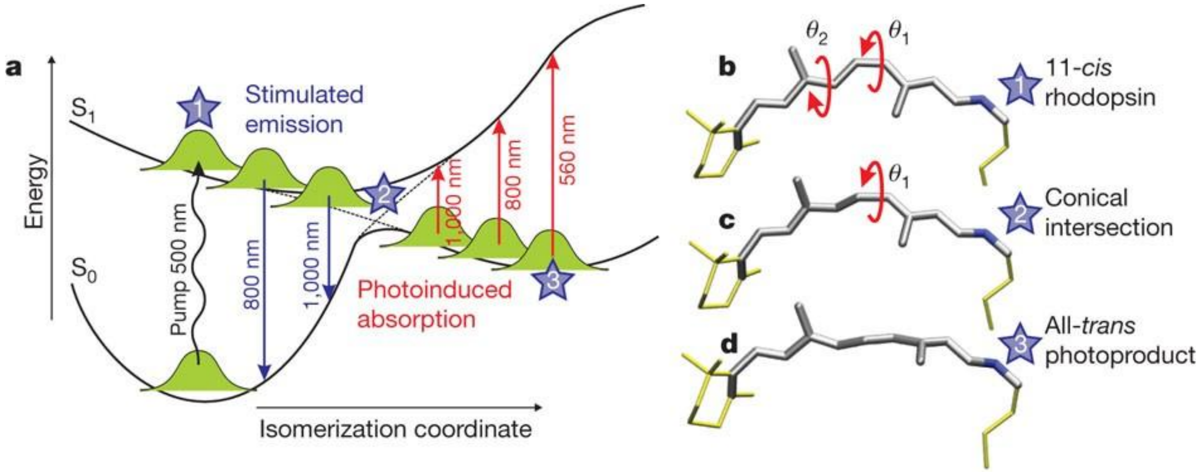
- Single-state LDR
- geometric phase manifested as a nodal line along the $\phi = 0$



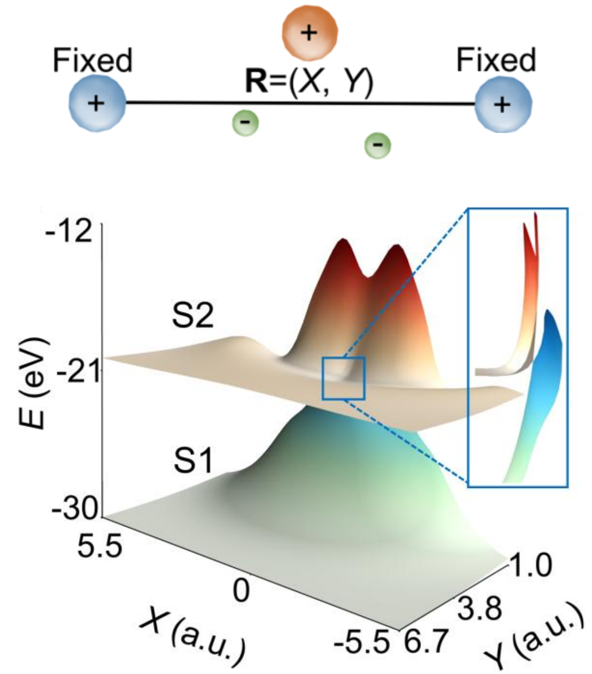


Lan et al., J. Chem. Phys. 122, 224315 (2005)
In preparation

The geometric phase in nonadiabatic dynamics is incorporated by LDR.



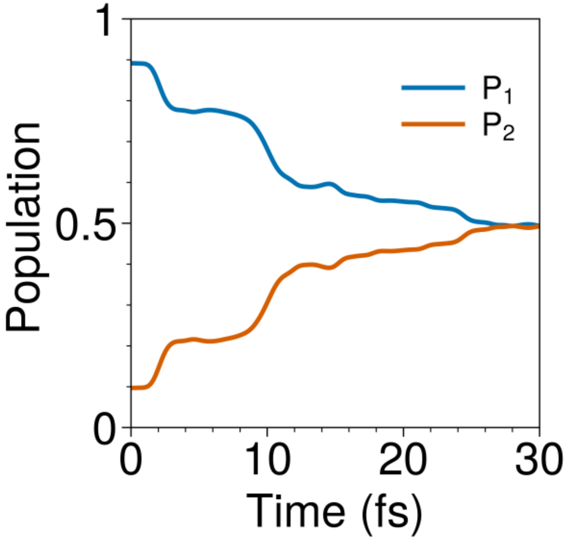
Polli et al. *Nature* 2010, *467* (7314), 440–443



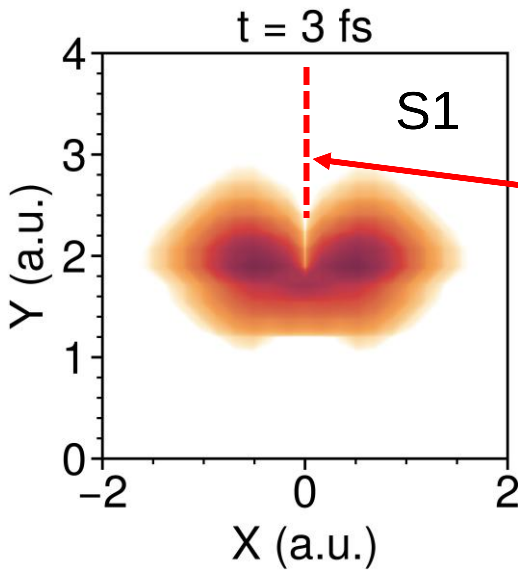
H_3^+

FCI/cc-pVTZ

Electronic nonadiabatic transition



Geometric phase effect



destructive interference

In preparation

Conclusions

- The local diabatic representation provides a numerically exact approach for exact quantum dynamic simulation without singularities.

- The single-state local diabatic representation enables the incorporation of the geometric phase effect using only a single potential energy surface.

- The local diabatic representation captures the geometric phase effect in nonadiabatic reactions.

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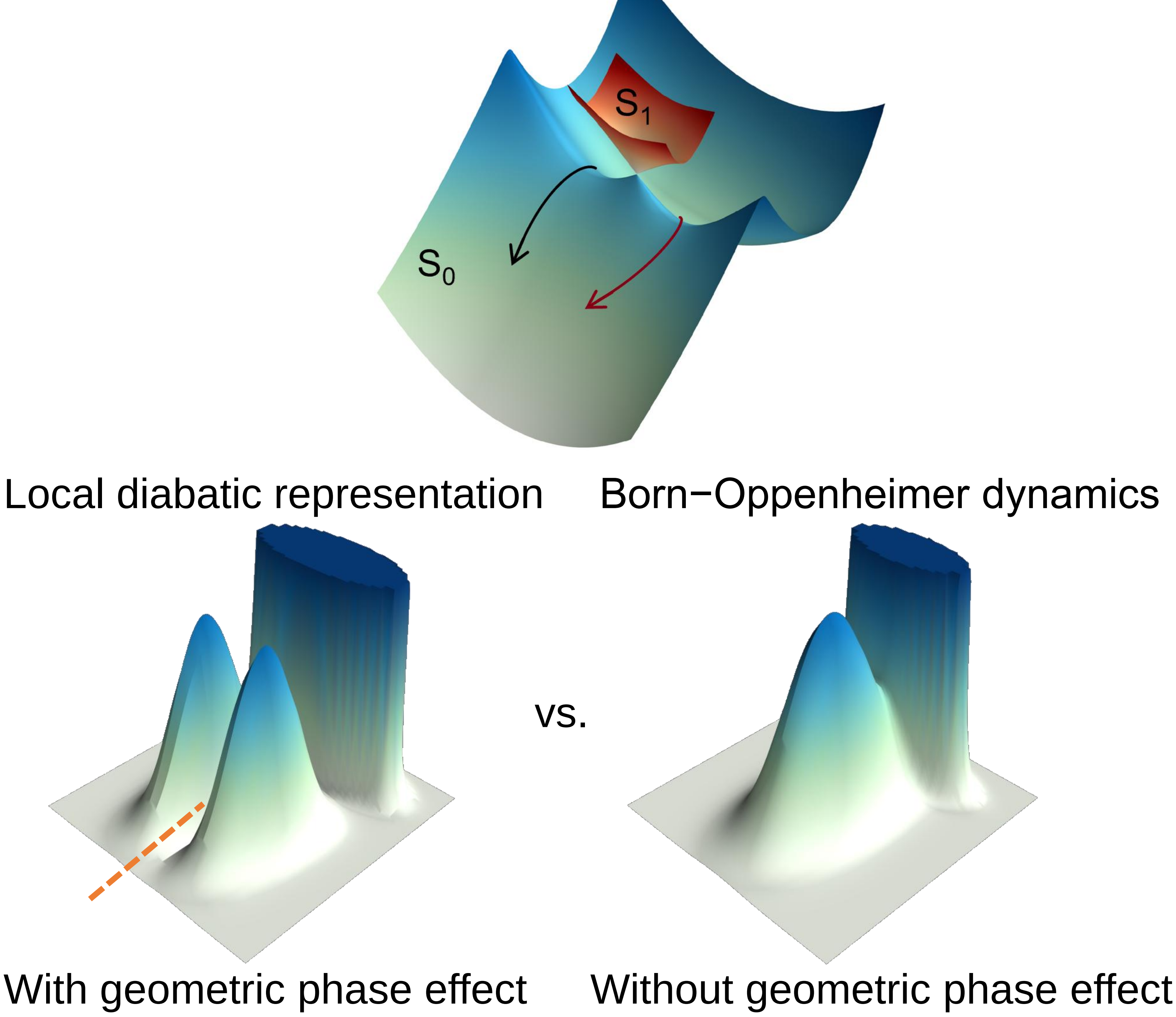
S1
S0
Local diabatic representation
Born−Oppenheimer dynamics
vs.
With geometric phase effect
Without geometric phase effect